Homework 8

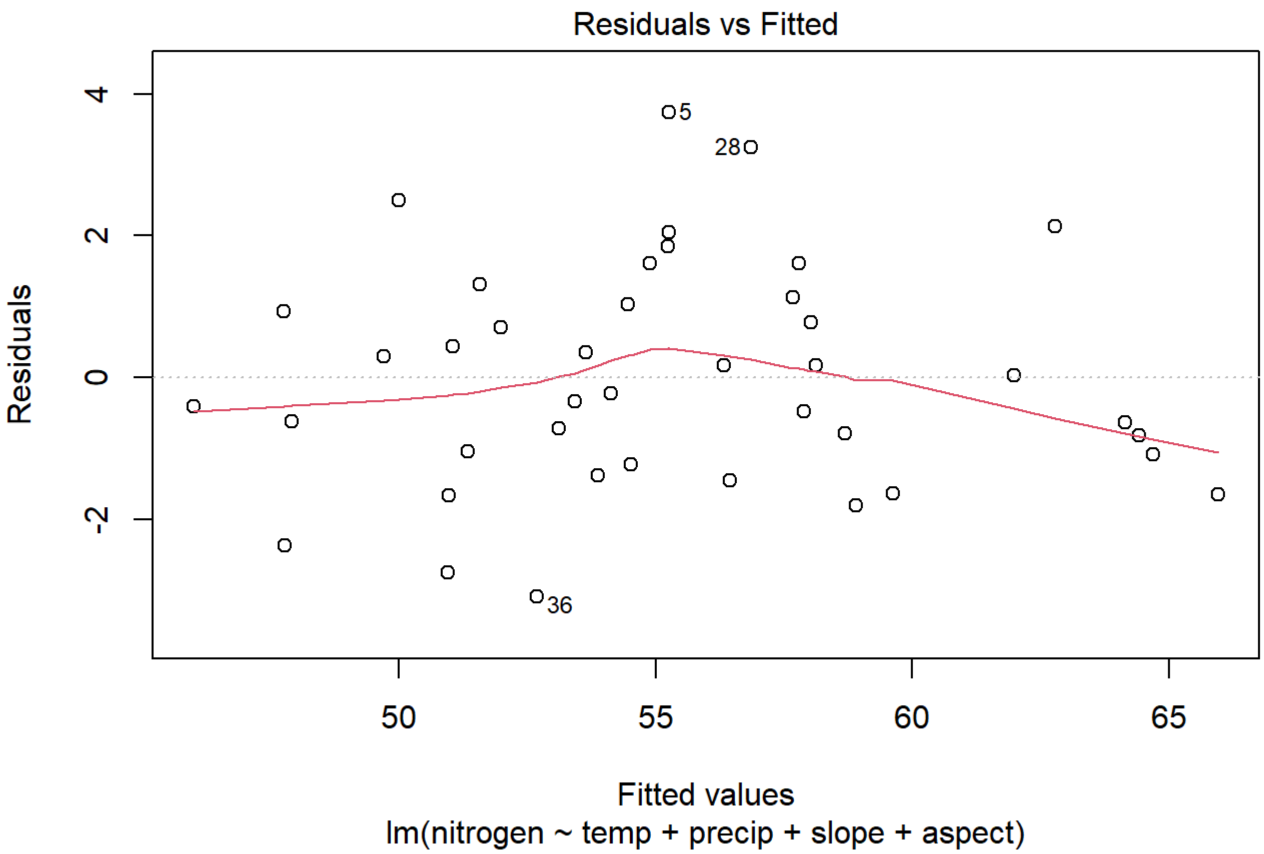
a

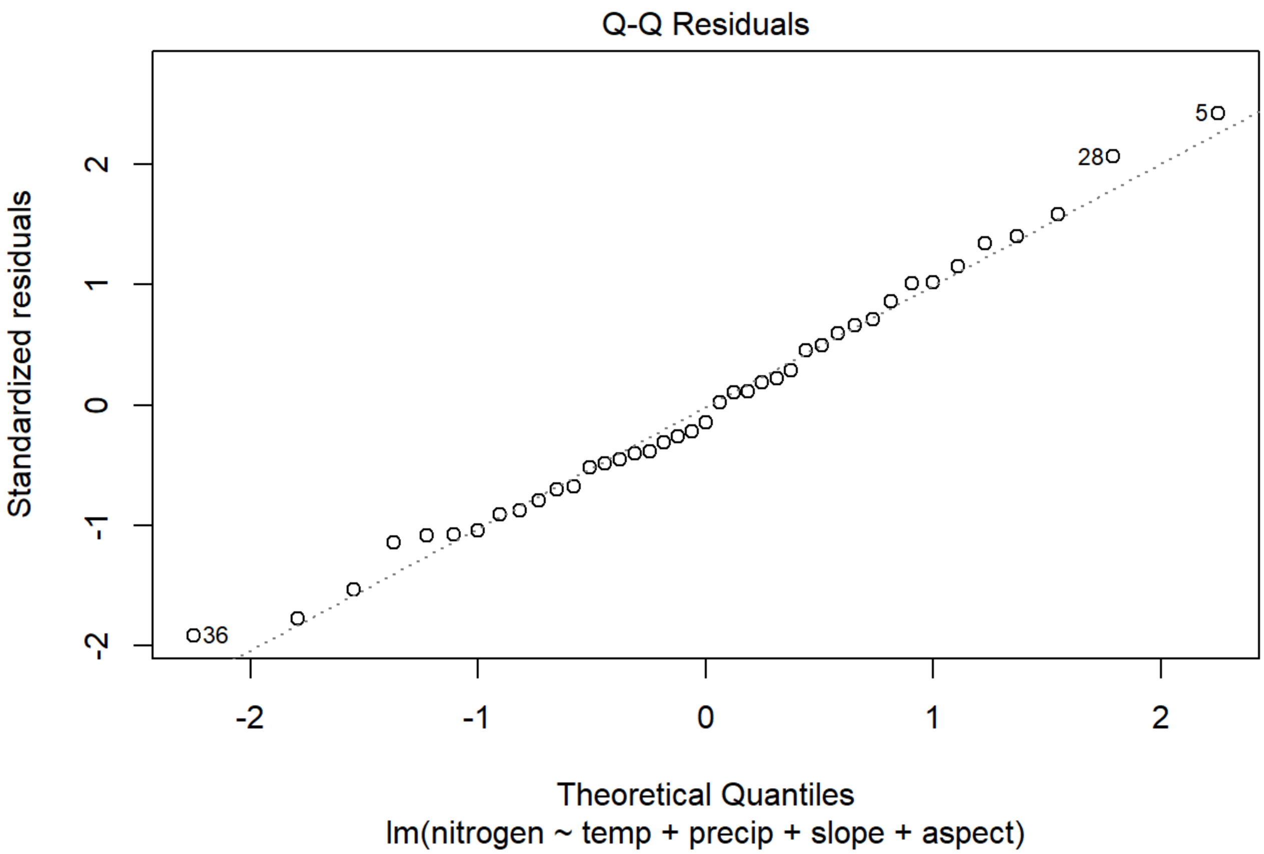
```r
soil <- read.csv("soil_nitrogen.csv")

soil$aspect <- factor(soil$aspect)

model <- lm(nitrogen ~ temp + precip + slope + aspect,
            data = soil)

summary(model)
```

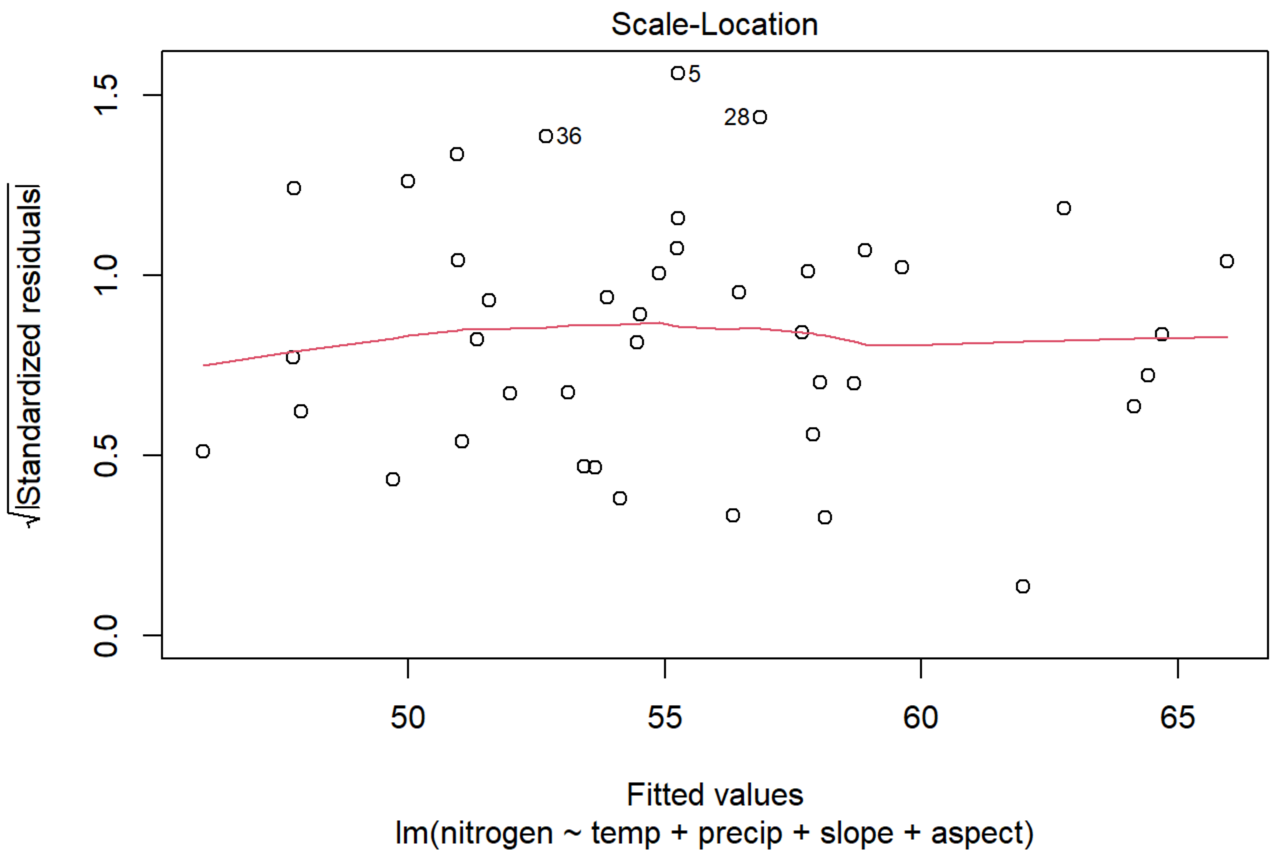
```
Call:
lm(formula = nitrogen ~ temp + precip + slope + aspect, data = soil)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.0914 -1.0880 -0.2175  1.0337  3.7394 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 47.63347    2.47280  19.263  < 2e-16 ***
temp         0.03736    0.09292   0.402   0.6900    
precip       0.11302    0.01054  10.726 9.23e-13 ***
slope       -0.78304    0.06029 -12.987 3.76e-15 ***
aspectS      1.25273    0.52804   2.372   0.0231 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.664 on 36 degrees of freedom
Multiple R-squared:  0.9075,	Adjusted R-squared:  0.8972 
F-statistic: 88.28 on 4 and 36 DF,  p-value: < 2.2e-16
```

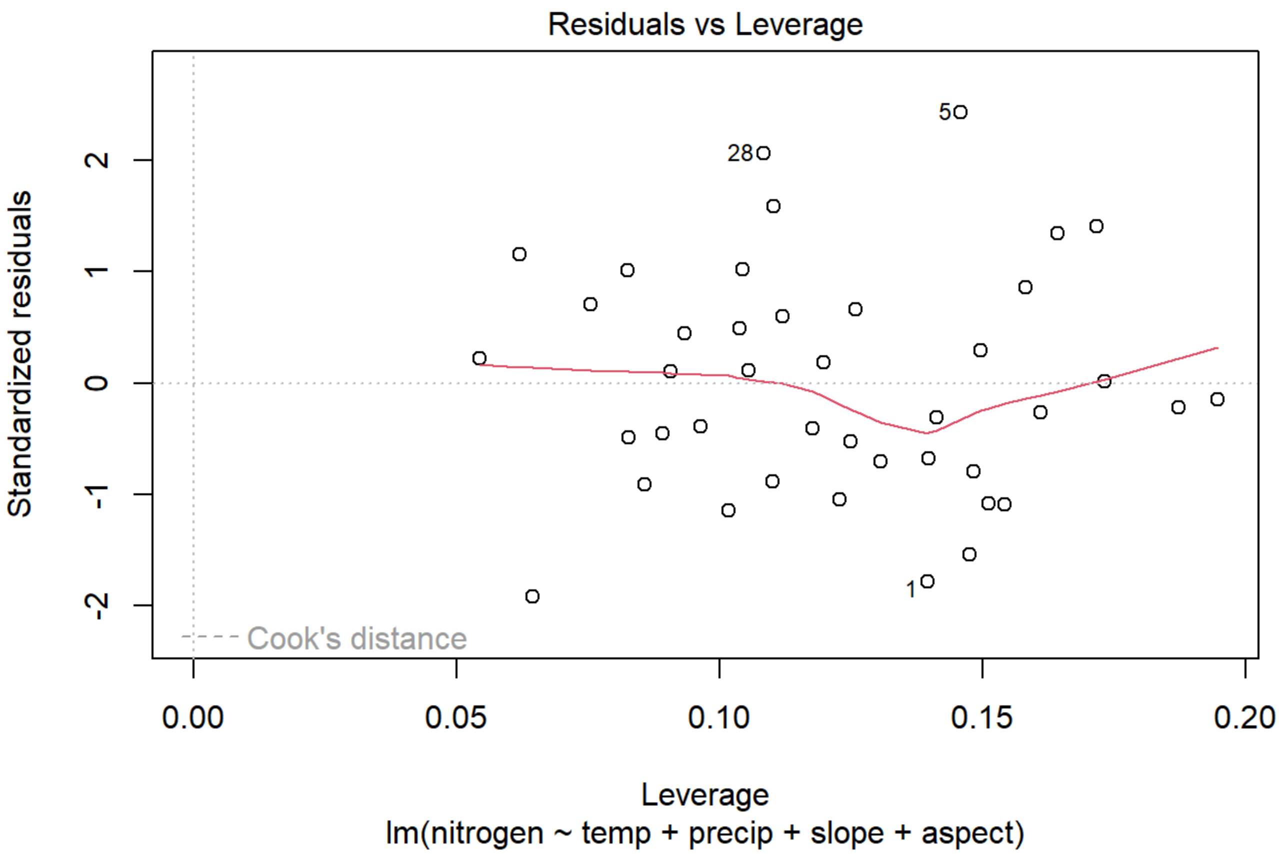
c

```r
plot(model)
```


Residuals vs Fitted
Residuals
4
2
0
-2
5
28
36
50
55
60
65
Fitted values
lm(nitrogen ~ temp + precip + slope + aspect)


Q-Q Residuals
Standardized residuals
2
1
0
-1
-2
5
28
36
-2
-1
0
1
2
Theoretical Quantiles
lm(nitrogen ~ temp + precip + slope + aspect)


Scale-Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
50
55
60
65
5
28
36
Fitted values
lm(nitrogen ~ temp + precip + slope + aspect)



d

- The residuals are normally distributed
- The Rsquared value is hight and explains most of the variation in the dependent variable.
- Most of the factors are significant except for temperature.

e

```
## install.packages("MuMIn)

library(MuMIn)
```

```
Warning: package 'MuMIn' was built under R version 4.5.3
```

```
# intercept
model1 <- lm(nitrogen ~ 1 , data = soil)
print("model1")
```

```
[1] "model1"
```

```
aic1 <- AIC(model1)
aicc1 <- AICc(model1)
bic1 <- BIC(model1)
aic1
```

```
[1] 254.3651
```

```
aicc1
```

```
[1] 254.6809
```

```
bic1
```

```
[1] 257.7923
```

```
## intercept + temp
model2 <- lm(nitrogen ~ temp , data = soil)
print("model2")
```

```
[1] "model2"
```

```
aic2 <- AIC(model2)
aicc2 <- AICc(model2)
bic2 <- BIC(model2)
aic2
```

```
[1] 256.3309
```

```
aicc2
```

```
[1] 256.9796
```

```
bic2
```

```
[1] 261.4717
```

```
## intercept + slope
model3 <- lm(nitrogen ~ slope , data = soil)
print("model3")
```

```
[1] "model3"
```

```
aic3 <- AIC(model3)
aicc3 <- AICc(model3)
```

```
bic3 <- BIC(model3)
aic3
```

```
[1] 221.0868
```

```
aicc3
```

```
[1] 221.7354
```

```
bic3
```

```
[1] 226.2275
```

```
## intercept + prep
model4 <- lm(nitrogen ~ precip , data = soil)
print("model4")
```

```
[1] "model4"
```

```
aic4 <- AIC(model4)
aicc4 <- AICc(model4)
bic4 <- BIC(model4)
aic4
```

```
[1] 232.4204
```

```
aicc4
```

```
[1] 233.069
```

```
bic4
```

```
[1] 237.5611
```

```
## intercept + aspect
model5 <- lm(nitrogen ~ aspect , data = soil)
print("model5")
```

```
[1] "model5"
```

```
aic5 <- AIC(model5)
aicc5 <- AICc(model5)
bic5 <- BIC(model5)
aic5
```

```
[1] 254.0923
```

```
aicc5
```

```
[1] 254.7409
```

```
bic5
```

```
[1] 259.233
```

```
## intercept + temp + slope
model6 <- lm(nitrogen ~ temp + slope , data = soil)
print("model6")
```

```
[1] "model6"
```

```
aic6 <- AIC(model6)
aicc6 <- AICc(model6)
bic6 <- BIC(model6)
aic6
```

```
[1] 222.3608
```

```
aicc6
```

```
[1] 223.472
```

```
bic6
```

```
[1] 229.2151
```

```
## intercept + temp + prep
model7 <- lm(nitrogen ~ temp + precip , data = soil)
print("model7")
```

```
[1] "model7"
```

```
aic7 <- AIC(model7)
aicc7 <- AICc(model7)
bic7 <- BIC(model7)
aic7
```

```
[1] 234.2511
```

```
aicc7
```

```
[1] 235.3622
```

```
bic7
```

```
[1] 241.1054
```

```
## intercept + temp + aspect
model8 <- lm(nitrogen ~ temp + aspect , data = soil)
print("model8")
```

```
[1] "model8"
```

```
aic8 <- AIC(model8)
aicc8 <- AICc(model8)
bic8 <- BIC(model8)
aic8
```

```
[1] 256.0258
```

```
aicc8
```

```
[1] 257.1369
```

```
bic8
```

```
[1] 262.8801
```

```
## intercept + slope + prep
model9 <- lm(nitrogen ~ slope + precip , data = soil)
print("model9")
```

```
[1] "model9"
```

```
aic9 <- AIC(model9)
aicc9 <- AICc(model9)
bic9 <- BIC(model9)
aic9
```

```
[1] 166.807
```

```
aicc9
```

```
[1] 167.9181
```

```
bic9
```

```
[1] 173.6613
```

```
## intercept + slope + aspect
model10 <- lm(nitrogen ~ slope + aspect , data = soil)
print("model10")
```

```
[1] "model10"
```

```
aic10 <- AIC(model10)
aicc10 <- AICc(model10)
bic10 <- BIC(model10)
aic10
```

```
[1] 220.4578
```

```
aicc10
```

```
[1] 221.5689
```

```
bic10
```

```
[1] 227.3121
```

```
## intercept + prep + aspect
model11 <- lm(nitrogen ~ precip + aspect , data = soil)
print("model11")
```

```
[1] "model11"
```

```
aic11 <- AIC(model11)
aicc11 <- AICc(model11)
bic11 <- BIC(model11)
aic11
```

```
[1] 232.1355
```

```
aicc11
```

```
[1] 233.2466
```

```
bic11
```

```
[1] 238.9898
```

```
## intercept + temp + prep + slope
model12 <- lm(nitrogen ~ slope + aspect , data = soil)
print("model12")
```

```
[1] "model12"
```

```r
aic12 <- AIC(model12)
aicc12 <- AICc(model12)
bic12 <- BIC(model12)
aic12
```

```
[1] 220.4578
```

```r
aicc12
```

```
[1] 221.5689
```

```r
bic12
```

```
[1] 227.3121
```

```r
## intercept + prep + slope + aspect
model13 <- lm(nitrogen ~ precip + slope + aspect , data = soil)
print("model13")
```

```
[1] "model13"
```

```r
aic13 <- AIC(model13)
aicc13 <- AICc(model13)
bic13 <- BIC(model13)
aic13
```

```
[1] 162.9527
```

```r
aicc13
```

```
[1] 164.667
```

```r
bic13
```

```
[1] 171.5205
```

```r
## intercept + temp + prep + aspect
model14 <- lm(nitrogen ~ slope + aspect , data = soil)
print("model14")
```

```
[1] "model14"
```

```r
aic14 <- AIC(model14)
aicc14 <- AICc(model14)
```

```
bic14 <- BIC(model14)
aic14
```

```
[1] 220.4578
```

```
aicc14
```

```
[1] 221.5689
```

```
bic14
```

```
[1] 227.3121
```

```
## intercept + temp + slope + aspect
model15 <- lm(nitrogen ~ slope + aspect , data = soil)
print("model15")
```

```
[1] "model15"
```

```
aic15 <- AIC(model15)
aicc15 <- AICc(model15)
bic15 <- BIC(model15)
aic15
```

```
[1] 220.4578
```

```
aicc15
```

```
[1] 221.5689
```

```
bic15
```

```
[1] 227.3121
```

```
## intercept + temp + prep + slope + aspect
model16 <- lm(nitrogen ~ slope + aspect , data = soil)
print("model16")
```

```
[1] "model16"
```

```
aic16 <- AIC(model16)
aicc16 <- AICc(model16)
bic16 <- BIC(model16)
aic16
```

```
[1] 220.4578
```

```r
aicc16
```

```
[1] 221.5689
```

```r
bic16
```

```
[1] 227.3121
```

f

```r
step_model <- step(model, direction = "both")
```

```
Start:  AIC=46.42
nitrogen ~ temp + precip + slope + aspect

         Df Sum of Sq    RSS     AIC
- temp    1      0.45 100.11  44.600
<none>                 99.66  46.416
- aspect  1     15.58 115.24  50.372
- precip  1    318.51 418.17 103.215
- slope   1    466.91 566.57 115.668

Step:  AIC=44.6
nitrogen ~ precip + slope + aspect

         Df Sum of Sq    RSS     AIC
<none>                100.11  44.600
+ temp    1      0.45  99.66  46.416
- aspect  1     15.37 115.47  48.454
- precip  1    327.24 427.35 102.105
- slope   1    468.06 568.17 113.783
```

```r
summary(step_model)
```

```

Call:
lm(formula = nitrogen ~ precip + slope + aspect, data = soil)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.0538 -1.0616 -0.3658  0.9503  3.8753 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 48.31030    1.79057  26.980  < 2e-16 ***
precip       0.11358    0.01033  10.998 3.23e-13 ***
slope       -0.78088    0.05937 -13.153 1.61e-15 ***
```

```
aspectS      1.24261    0.52143   2.383   0.0224 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.645 on 37 degrees of freedom
Multiple R-squared:  0.9071,    Adjusted R-squared:  0.8995
F-statistic: 120.4 on 3 and 37 DF,  p-value: < 2.2e-16
```

g

Y = Bo + B5X5 + B2X2 + B3X3 + B4X4

```
Y = 47.633 + (0.03736*22) + (0.55302*550) + (-0.78304*55) + (5.25273*5)
Y
```

```
[1] 335.8124
```

Based on the model, the value of nitrogen concentration is 58.04725